

■ 3.6.4 Advanced Topic: Composition and Inversion of Power Series

When you manipulate power series, it is sometimes convenient to think of the series as representing *functions*, which you can, for example, compose or invert.

<code>series₁ /. x->series₂</code>	compose two power series
<code>InverseSeries[series]</code>	invert a power series

Composition and inversion of power series.

Here is the power series for $\sin(x)$ to $O(x^5)$.

```
In[1]:= Series[Sin[x], {x, 0, 5}]
```

$$\text{Out}[1]= x - \frac{x^3}{6} + \frac{x^5}{120} + O[x]^6$$

The variable in the original power series is now replaced by the power series for $\sin(x)$.

```
In[2]:= % /. x -> Series[Sin[x], {x, 0, 5}]
```

$$\text{Out}[2]= x - \frac{x^3}{3} + \frac{x^5}{10} + O[x]^6$$

The result is the power series for $\sin(\sin(x))$.

```
In[3]:= Series[Sin[Sin[x]], {x, 0, 5}]
```

$$\text{Out}[3]= x - \frac{x^3}{3} + \frac{x^5}{10} + O[x]^6$$

If you think of a power series in x as approximating a function $f(x)$, then substituting a power series for another function $g(x')$ gives a power series for the composed function $f(g(x))$. This power series can only be constructed if the first term in the power series for the inner function $g(x)$ involves a positive power of x .

If you have a power series for a function $f(x)$ where the first term is proportional to x , then it is possible to find the power series for the inverse function $f^{-1}(x)$ such that $f(f^{-1}(x)) = x$. The operation of finding the power series for an inverse function is sometimes known as *reversion* of power series.

Here is the series for $\sin(x)$.

```
In[4]:= Series[Sin[x], {x, 0, 5}]
```

$$\text{Out}[4]= x - \frac{x^3}{6} + \frac{x^5}{120} + O[x]^6$$

Inverting the series gives the series for $\sin^{-1}(x)$.

```
In[5]:= InverseSeries[%]
```

$$\text{Out}[5]= x + \frac{x^3}{6} + \frac{3x^5}{40} + O[x]^6$$

Composing the two series gives the identity function.

```
In[6]:= % /. x -> %%
```

$$\text{Out}[6]= x + O[x]^6$$